ROS 2 Fundamentals — Environment Checklist

Module 0 · Course Onboarding and Roadmap

HARDWARE & OS

- A 64-bit machine with 8GB+ RAM (16GB recommended if running Ubuntu inside a VM)
- 25GB+ free disk space
- A GPU capable of basic OpenGL 3.3+ — integrated graphics from the last ~8 years is generally fine. This matters for Gazebo in Module 14, not for anything before it.
- Ubuntu 24.04 LTS ("Noble Numbat") — native, dual-boot, VM, or WSL2 are all valid setups; trade-offs are covered honestly in Module 3.

BACKGROUND & READINESS

- [ ] Comfortable using a terminal (running commands, navigating directories)
- [ ] Basic programming literacy in at least one language
- [ ] Willingness to troubleshoot — this course teaches debugging as a real skill, not an afterthought

> Missing one of these is fine. Module 3 teaches ROS 2-specific terminal use from scratch — general comfort is enough, prior ROS experience is not required.

LANGUAGE STRATEGY

This course teaches **Python** as the primary path throughout. C++ examples appear occasionally to show that ROS 2 concepts transfer across languages — never as a second full track you need to follow in parallel.

VERSION PIN (THIS COURSE TARGETS EXACTLY THIS STACK)

- ROS 2 Distribution: **Jazzy Jalisco** (LTS)
- Ubuntu: **24.04 LTS**
- Simulator: **Gazebo Harmonic** — not Gazebo Classic, which reached end-of-life and does not run on Ubuntu Noble at all
- Python: **3.12**
- C++ Standard: **C++17**